

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with **Protocol Buffers** – orchestrated communication between Chrome components at Google's security review standard; evaluated Protobuf vs. custom serialization (schema evolution, memory safety, cross-language compatibility); shipped to Chrome stable serving **3B+ active users** at **sub-50ms p99; 95% test coverage**; zero production regressions
- Architected event-driven **TypeScript/React** system with observer pattern across **25K+ lines** of Chromium – decoupled UI state propagation from component hierarchy; **68% feature delivery acceleration** at **95% test coverage**; design documents and code reviews adopted into production branch by senior Chrome engineers
- Profiled settings navigation at 1,200ms p99; designed and implemented **lock-free concurrent trie search** with **CAS operations** eliminating mutex contention – **96% latency reduction**, zero production regressions

TiMoto AI | Software Engineer

Sep 2025 – Present

- Designed **gRPC inter-service layer** (Go/Python) with exactly-once delivery – traced production deadlock to shared resource acquisition conflicts under concurrent load; redesigned call sequencing; **100% evaluation success rate** at **sub-50ms p99**; primary engineer on 3-person team owning backend, infra, and ML serving end-to-end
- Architected **multi-AZ AWS ECS Fargate** with **Terraform IaC**, circuit breaker pattern, and auto-rollback on health check failure – **99.9% uptime SLO**; CloudWatch + Prometheus/Grafana observability; on-call rotation, root-cause analysis, runbooks, post-mortem; **44% infrastructure cost reduction** (\$40-60/mo)
- Built **LLM-as-a-judge evaluation pipeline** – automated quality gate before deployment; **LangChain** orchestration + LLM API integration (Claude, GPT); replaced manual review with automated regression detection; zero production regressions
- Deployed **vLLM with PagedAttention** for production ML serving – solved KV cache memory fragmentation under concurrent load; **zero OOM failures**; continuous batching throughput optimization

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless **BaaS on AWS** – selected **JWT** over session auth for stateless horizontal scalability; auto-scaling supporting **500+ concurrent users**; **90% deployment time reduction** via CI/CD (**GitHub Actions**)
- Diagnosed N+1 query bottleneck degrading response to 3s+; redesigned with **PostgreSQL composite indexing** achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | *Go, TypeScript, Multi-Cloud IaC*

github.com/pulumi

- Submitted **Go CLI features** for multi-cloud (AWS/Azure/GCP) provisioning – open-source developer tooling under active review by core maintainers; 24.4K+ star project
- Analyzed **Raft/Paxos consensus** in distributed state synchronization – studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

Languages: C++, TypeScript, Go, Python, JavaScript, Rust, Java, SQL, Bash

Frameworks & UI: React, FastAPI, Django, Node.js, REST APIs, PostgreSQL, Redis, MongoDB, HTML/CSS

Distributed Systems: gRPC, Protocol Buffers, Exactly-once semantics, Circuit breakers, Multi-AZ, Fault tolerance, Event-driven architecture

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, Kubernetes, Docker, CI/CD (GitHub Actions), CloudWatch, Prometheus, Grafana

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LLM-as-a-judge evaluation, LangChain, PyTorch, LLM APIs (Claude, GPT)

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor